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Asymmetric Numeral System

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Outline

Introduction

Introduction

- Hi there, this is a self inspired latex beamer template for Leipzig university.

Entropy Coding

1st frame

- This is one of the section[1].
- Let's check for 2nd footnote [2].
- The footnotes will be added from the bottom. If you want to pull the footnotes towards red-black margin line then use `\addfootspace{}` at the end of the `\frame{}`.
- `\addfootspace{}` actually adds an additional vacant footnote (without counting that vacant footnote).
- To add more foot space add more `\addfootspace{}`.

[1] This is a long footnote. Here I am. Who am I?

[2] This is a 2nd long footnote. Here I am. Who am I?

2nd frame

- Here is the next section[3]
- Here I used two `\addfootspace{}`. Don't do `\addfootspace{\vspace{1cm}}`.

[3] This is a 2nd long footnote. Here I am. Who am I?

Asymmetric Numeral System

Motivation

ANS combines the compression ratio of arithmetic coding (which uses a nearly accurate probability distribution), with a processing cost similar to that of Huffman coding [3]

Basic Idea

– Symbol spread function

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- Representation of information
  (represents an entire message as a single number, similar to Arithmetic Coding) –> example: representation of {2,0,2,5,1,8}

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- Symmetric or Asymmetric
- rANS and tANS

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Symbol spread function

- the function that determines the split of the $\mathbb{N}$ subsets corresponding to different symbols [1]
- $\bar{s} : \mathbb{N} \to \mathcal{A}, \bar{s}(x) = \mathrm{mod}(x, b) = s$
- In the standard binary system example: $\bar{s}(x) = \mathrm{mod}(x, 2)$
  e.g. splits natural numbers into subsets of even/odd numbers [1]

Symbol spread function (Exa. 1)

$S = \{0, 1\}$

Exa. 1: Standard Binary System

N	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
s=0																			
s=1																			

Fig.1

Symbol spread function (Exa. 2)

$S = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

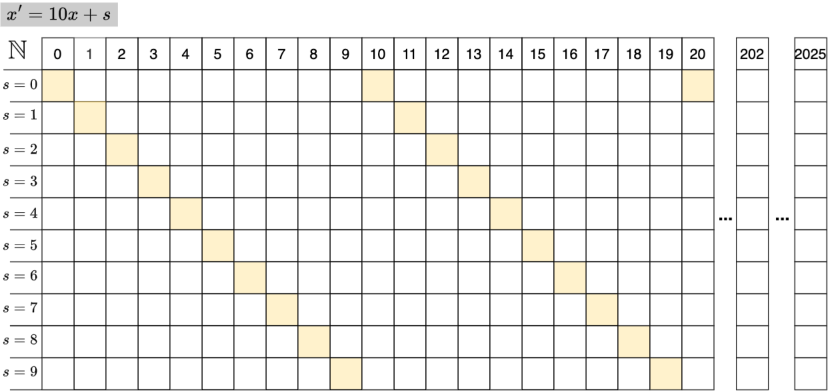

Fig.2

Representation of information (Exa. 2)

- a finite sequence of symbols/digits:

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- How can we represent this sequence as a single number?
- The easiest way is to concatenate the symbols/digits:

$$x_{out} = 2025$$

Representation of Information (Exa. 2)

- encoding the sequence of digits (2,0,2,5,1,8)

$$
\begin{aligned}
x_{\text{init}} &= 0 \\
x_1 &= x_{\text{init}} \times 10 + 2 = 2 \\
x_2 &= x_1 \times 10 + 0 = 20 \\
x_3 &= x_2 \times 10 + 2 = 202 \\
x_{\text{out}} &= x_3 \times 10 + 5 = 2025
\end{aligned}
$$

Representation of Information (Exa. 2)

– What is the encoding function?

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$$x_{i+1} = x_i \times 10 + s_i$$

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$$x_{i+1} = x_i \times 10 + s_i$$

- x' is the x-th appearance of symbol s

$$x' = C(s, x) = x \cdot 10 + s$$

Representation of Information (Exa. 2)

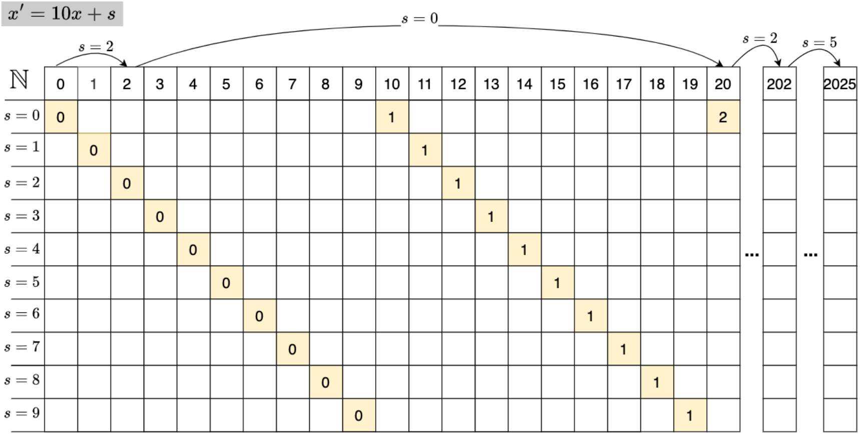

Fig.3

Representation of Information (Exa. 1)

Exa. 1: Standard Binary System

$X' = 2X + s$

s=1 s=0 s=0 s=1

		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
X'		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
X	s=0	0		1		2		3		4		5		6		7		8		9
	s=1		0		1		2		3		4		5		6		7		8	

Fig.4

Exa. 1: Standard Binary System

X' = 2X + s

s=1 s=0 s=0 s=1

x'		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
x	s=0	0		1		2		3		4		5		6		7		8		9
	s=1		0		1		2		3		4		5		6		7		8	

Exa. 2: Standard Decimal System

X' = 10X + s

s=2 s=0 s=2 s=5

N	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		202		2025
s=0	0										1										2				
s=1		0										1													
s=2			0										1												
s=3				0										1											
s=4					0										1							...		...	
s=5						0										1									
s=6							0										1								
s=7								0										1							
s=8									0										1						
s=9										0										1					

Fig.5

Asymmetric Numeral System | Basic Idea | Symmetric or Asymmetric

Exa. 1: Standard Binary System

X' = 2X + s

s=1 s=0 s=0 s=1

		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
X'		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
X	s=0	0		1		2		3		4		5		6		7		8		9
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Exa. 2: Standard Decimal System

X' = 10X + s

s=2 s=0 s=2 s=5

N	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		202		2025
s=0	0										1										2				
s=1		0										1													
s=2			0										1												
s=3				0										1											
s=4					0										1							...		...	
s=5						0										1									
s=6							0										1								
s=7								0										1							
s=8									0										1						
s=9										0										1					

Fig.6

Exa. 3: Asymmetric Binary System

- Let $\mathcal{A} = \{0, 1\}$
- $Pr(0) = 1/4, Pr(1) = 3/4$

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- $x' = C(s, x) \approx x/p_s$

Exa. 3: Asymmetric Binary System

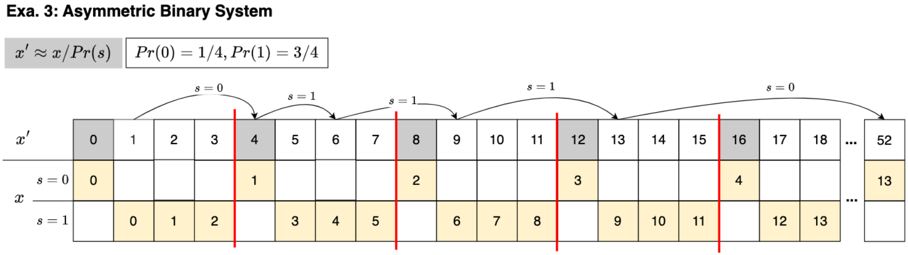


Fig.7

We refine even/odd number by changing their densities

Symmetric or Asymmetric

- Symmetric: uniform distribution
- Asymmetric: non-uniform distribution

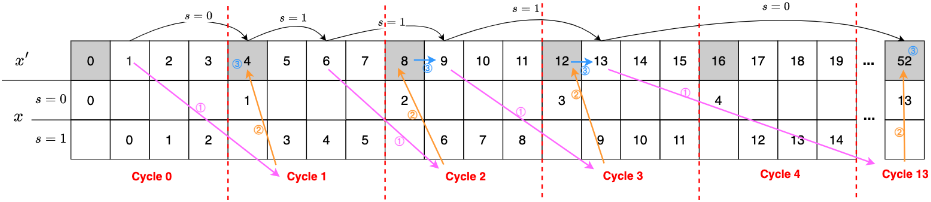
rANS

- ”r” = ”range”: similar to range coding (RC)
- symbol spread function
- representation of information
- asymmetric

Exa.3

How to encode 101110? What is the encoding function $C(s, x)$?
$x' = C(s, x) \approx x/p_s$

$x' \approx x/Pr(s)$ $\quad$ $Pr(0) = 1/4, Pr(1) = 3/4$



Given current state x and next symbol s. How to find next state x', which is the x-th appearance of symbol s ?

	$s=0$	$s=1$	0 and 1
Step ①: Find cycle number k	$k=\lfloor x/1 \rfloor = x$	$k = \lfloor x/3 \rfloor$	$k=\lfloor x/F_s \rfloor$ F_s:
Step ②: Navigate to the beginning (first number) of that cycle	$4 \cdot k$	$4 \cdot k$	$m \cdot k$ m: size of the cycle
Step ③: Find the position of the symbol s in that cycle	part 1: first of one + 0	part 1: + $\mathrm{mod}(x, 3)$	part 1: + $\mathrm{mod}(x, F_s)$
- part 1: position of the symbol s in the sequence of the same symbol - part 2: how many other symbols before the symbol s	part 2: no other symbols + 0	part 2: only one other symbol (0) and with one of that symbol in one cycle + 1	part 2: only one other symbol (0) and with one of that symbol in one cycle $+c_s$
	$x' = 4 \cdot x$	$x' = 4 \cdot \lfloor x/3 \rfloor + \mathrm{mod}(x,3) + 1$	$x' = m \cdot \lfloor x/F_s \rfloor + \mathrm{mod}(x, F_s) + c_s$

$$x' = C(s, x) = m \cdot \lfloor x/F_s \rfloor + \mathrm{mod}(x, F_s) + c_s$$

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Fig.8

Encoding Function (general):

$$x' = m \cdot \left\lfloor \frac{x}{F_s} \right\rfloor + \mathrm{mod}(x, F_s) + c_s$$

- **Alphabet set:** $\mathcal{A} = \{a_1, a_2, \ldots, a_k\}$
- **Absolute frequency of symbol** s**:** F_s
- **Total frequency:** $m = \sum_{i=1}^{k} F_i$
- **Probability of symbol** s**:** $p_s = \frac{F_s}{m}$
- **Cumulative frequency before** s**:** $c_s := F_0 + F_1 + \cdots + F_{s-1}$

Encoding Function (binary):

$$x' = C(s, x) = \left\lfloor \frac{x}{F_s} \right\rfloor << n + \mathrm{mod}(x, F_s) + c_s \; [1]$$

- **Alphabet set:** $\mathcal{A} = \{a_1, a_2, \ldots, a_k\}$
- **Absolute frequency of symbol** s**:** F_s
- **Total frequency:** $m = \sum_{i=1}^{k} F_i$
- **Probability of symbol** s**:** $p_s = \frac{F_s}{m}$
- **Cumulative frequency before** s**:** $c_s := F_0 + F_1 + \cdots + F_{s-1}$
- $<<$ n is a left shift by n bits

Decoding Function (binary):

$$D(x) = (s, F_s \cdot (x >> n) + (x \& mask) - c_s)^{[1]}$$

Implementation

- **Python Implementation:**

Streaming-rANS

To avoid using large number arithmetic, in practice stream variants are used: which enforce by renormalization: sending the least significant bits of x to or from the bitstream (usually L and b are powers of 2).

tANS

- d

Performance

- f

Summary

Summary

- To summarize, this may help you to make a Leipzig university beamer presentation.

Reference

Reference

1. J. Duda, K. Tahboub, N. J. Gadgil and E. J. Delp, ”The use of asymmetric numeral systems as an accurate replacement for Huffman coding,” 2015 Picture Coding Symposium (PCS), Cairns, QLD, Australia, 2015, pp. 65-69, doi: 10.1109/PCS.2015.7170048.
2. J. Duda, “Asymmetric numeral systems: entropy coding combining speed of huffman coding with compression rate of arithmetic coding,” arXiv:1311.2540.
3. Asymmetric numeral systems - Wikipedia
4. What is Asymmetric Numeral Systems? - Kedar Tatwawadi